

Sammy Sharief

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Education

Université Paris-Saclay

PhD in Physics

- Advisors: François Lanusse, Tobias Liaudat, and Samuel Farrens

Dec 2025 – Current

University of Montréal

Master of Science in Computer Science

- Advisor: Laurence Perreault-Levasseur

Sept 2023 – Sept 2025

University of Illinois at Urbana-Champaign

*Bachelor of Science in Liberal Arts & Sciences in Computer Science + Astronomy
Minor in Informatics*

Aug 2019 – May 2023

- Advisor: Gautham Narayan

Experience

MIRA: A Score for Conditional Distribution Accuracy and Model Comparison

PhD Project

ICML 2026 Main Conference Spotlight Paper

- Spearheaded MIRA, a novel metric for evaluating conditional distribution accuracy and enabling model comparison across domains
- Validated MIRA across modalities and high-dimensional settings, demonstrating reliable model comparison through extensive experiments
- Led collaboration efforts by communicating results, overseeing paper drafts, and maintaining a well-documented, user-friendly repository

Paris, France

Oct 2024 – April 2026

PQMass: Probabilistic Assessment of the Quality of Generative Models using Probability Mass Estimation

Master's Project

- Co-developed PQMass, a statistical framework that quantifies generative model performance and tests whether two distributions match
- Benchmarked PQMass across modalities and dimensions, exposing failure modes of common metrics such as FID and FLD
- Maintained the PQMass repository, ensuring robust performance, scalability to millions of dimensions, and handling of complex edge cases

Montréal, QC, CA

Feb 2024 – Jan 2025

DECam Data Reduction

Research Assistant

- Ran data reduction for the Dark Energy Camera (DECam) supporting the Young Supernova Experiment and external DECam users
- Improved pipeline efficiency, built supernova target templates, and processed member requests for proposals and publications
- Authored detailed pipeline documentation covering reduction workflows and the improvements I contributed

Champaign, IL, USA

May 2023 – Aug 2023

Learning SESNA with Machine Learning

Research Experience for Undergraduate NSF Intern

- Applied Multiple Imputations with Chained Equations to recover missing data across 9M Spitzer Extended Solar Neighborhood Archive (SESNA) sources
- Upsampled SESNA to balance class representation prior to training a Multi-Layer Perceptron (MLP)
- Classified 2.2M of 4.5M unlabeled sources using a probabilistic MLP classifier

Austin, TX, USA

June 2022 – Aug 2022

Light Deflection near Kerr-Newman black hole spacetime

Research Experience for Undergraduate NSF Intern

Rochester, NY, USA

June 2021 – Aug 2021

- Investigated equatorial-plane photon trajectories near Kerr-Newman black holes, parameterized by mass, spin, and charge
- Derived approximate forms for photon orbits coinciding with the Innermost Circular Orbit as a function of the black hole's parameters
- Organized post-internship meetings to drive continued progress, culminating in a poster presentation at AAS 241

The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae

Research Student

Champaign, IL, USA

Jan 2021 – Apr 2023

- Built a pipeline to ingest Zwicky Transient Facility data and merge overlapping sources with the Young Supernova Experiment, producing an in-depth observational dataset
- Verified simulated supernovae against observations by extracting features with a modified Bazin fit, reducing dimensions with PCA, and using a KD-Tree for similarity matching

Publications

MIRA: A Score for Conditional Distribution Accuracy and Model Comparison [↗](#)

May 2026

Sammy Sharief, et al. ICML 2026 Main Conference Spotlight Paper

Uncertainty quantification metrics for scientific application [↗](#)

September 2025

Sammy Sharief Master's Thesis, University of Montréal

Pokie: Posterior Accuracy and Model Comparison

June 2025

Sammy Sharief, et al. Accepted to ML4Astro ICML 2025 Workshop

Field-Level Comparison and Robustness Analysis of Cosmological N-Body Simulations [↗](#)

May 2025

Adrian Bayer, Francisco Villaescusa-Navarro, **Sammy Sharief**, et al. Submitted to AAS

PQMass: Probabilistic Assessment of the Quality of Generative Models using Probability Mass Estimation [↗](#)

Jan 2025

Pablo Lemos, **Sammy Sharief**, et al. Accepted to ICLR Main Conference 2025

Searching for Bumps in the Cosmological Road: Do Type Ia Supernovae with Early Excesses Have Biased Hubble Residuals? [↗](#)

Feb 2024

Christine Ye, David O. Jones, Willem B. Hoogendam, Benjamin J. Shappee, Suhail Dhawan, **Sammy Sharief**
Accepted by ApJ

The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae [↗](#)

Feb 2023

Patrick Aleo, Konstantin Malanchev, **Sammy Sharief**, et al. Accepted by ApJ

SN 2022ann: A type Icn supernova from a dwarf galaxy that reveals helium in its circumstellar environment [↗](#)

Nov 2022

Kyle W Davis, et al Accepted by ApJ

Talks and Posters

NSF Funded SkAI Institute, Uncertainty Quantification for Generative Models in Astrophysics, *Talk*

Chicago

Feb 2026

International Conference on Learning Representations (ICLR), PQMass: Probabilistic Assessment of the Quality of Generative Models using Probability Mass Estimation, *Poster*

Singapore

April 2025

Centre de recherche en astrophysique du Québec, PQMass and its applicability to Astrophysics, *Talk*

Quebec, Canada

May 2024

Astronomy Festival (AstroFest 2023), Learning SESNA with Machine Learning, *Poster*

Illinois, USA

Jan 2023